Pseudocode-Map Substitution Grows with Expertise

Razieh Fathi and Ian Handy

Abstract—This study examined how novice and experienced programmers attended to Pseudocode and maps presented simultaneously while viewing video tutorials about breadth-first search (BFS) and depth-first search (DFS) graph traversals. Using eye tracking technology, we recorded where each of our 62 participants directed their gaze toward the Pseudocode or the Map during each of the 117 trials they completed. As programming knowledge increased from novice to intermediate to advanced, the correlations between how much time was spent gazing at the Pseudocode versus the Map became significantly more negative. This represents a type of Expertise Reversal Effect. While there was a similar trend in terms of how many times each area of interest was viewed, the trend did not reach statistical significance. Therefore, although these findings support the idea that a design intended to be optimal for novice learners might be suboptimal for expert learners, they also suggest caution regarding generalizing results based upon findings obtained from a limited number of novice learners.

Index Terms—eye-tracking, algorithm visualization, expertise reversal effect, dual representation, representation substitution

1 Introduction

The ability to combine multiple representations has been shown to enhance learning [4]. However, the extent to which such combinations are effective will depend upon the prior experiences of those who receive instruction. Both Cognitive Load Theory (CLT) [7] and the Expertise Reversal Effect (ERE) [3] propose that instructional aids, which are beneficial for novices, do not provide additional benefits for experienced learners. CLT states that the primary source of cognitive load is due to the complexity of the content being learned. When an experienced learner receives instruction via an aid designed for novices, ERE suggests that the added load created by the aid will outweigh its potential benefits. ERE proposes that as learners gain experience and form a schema related to the subject matter being taught, aids that previously supported the learning process become unnecessary.

Therefore, in order to determine whether such a trade-off occurs and whether the trade-off varies depending upon the level of expertise of a given learner, we calculated the Spearman correlation coefficient between the total amount of time spent gazing at the Pseudocode and the total amount of time spent gazing at the Map. These calculations allowed us to assess the degree to which learners substituted one representation for another.

Contributions: (1) we found significant evidence that the amount of time experienced learners spend gazing at Pseudocode and/or a Map decreases as they begin to substitute one for the other; (2) we provided a correlation-based method for quantifying representation substitution; and (3) we suggested implications for designing split-panel visualizations that cater specifically to certain levels of expertise.

2 Literature Review

CTML (Cognitive Theory of Multimedia Learning), developed by Mayer [4], posits that presenting verbal and pictorial information via separate pathways into working memory allows for better integration than when both types of information are combined. Visualizations involving split panels apply this concept directly to the presentation of algorithms. CLT provides a theoretical basis for understanding why adding a second representation to an existing representation would increase cognitive load. Specifically, CLT states that when the intrinsic difficulty of a task is already high, a secondary representation will result in greater cognitive load than a single representation [7, 8]. Finally, ERE expands on CLT by proposing that instructional aids that assist novices become redundant once learners have formed a schema related to the subject matter being studied [3].

A meta-analysis conducted by Gegenfurtner [2] revealed that intermediate learners tend to focus more than both novice and expert learners, which challenges the traditional view that as prior knowledge increases so too does focused attention. Bednarik [1] demonstrated that experts and novices differ in their attentional strategies when debugging algorithms. Experts verify their mental models whereas novices oscillate back-and-forth between panels without relying upon a strategic approach to problem solving. Similarly, Richter and Scheiter [6] discovered that low-prior-knowledge learners use signals to coordinate text and picture areas-of-interest (AOI) when providing cues; however, high-prior-knowledge learners do not benefit from cues.

Neither Bednarik nor Richter and Scheiter tested three levels of expertise. Furthermore, neither study measured correlation between dwell times as a metric. To demonstrate a substitution gradient requires both measures.

3 Methods

3.1 Using METAL as an Interactive Algorithm Visualization

Transfer of learning occurs when learners apply previously acquired knowledge to new problems or contexts [5]. Moving beyond surface-level schemas often requires additional instructional support, so that learners can attend to the reasoning behind each step of a solution rather than its form alone. Interactive visualizations embedded in example-based hypermedia have been suggested as one possible support [10].

We use the Map-based Educational Tools for Algorithmic Learning (METAL) project [9] as our visualization platform. METAL renders algorithm execution on Leaflet-based geographic maps alongside color-coded data-structure tables, and has been reported to increase student engagement in coursework that uses it [9]. For the present study we use METAL's visualizations for breadth-first search (BFS) and depth-first search (DFS) applied to a single map dataset—the Delaware road network (henceforth DE; 735 vertices, 879 edges; August 2022 TMG snapshot). Figure 1 shows a representative frame from one trial.

3.2 Design of the Study

The study employed a two-factor between/within design. The between-subject factor was prior programming experience (three groups: novice, intermediate, advanced; see Section 3.3). The within-subject factor was algorithm (BFS, DFS); every participant viewed both algorithms in the same session. Both BFS and DFS are standardly covered in introductory data-structures curricula, and are discoverable, if not always comprehended, by computer-science majors and non-majors alike. Trial order and other counterbalancing of BFS vs. DFS presentation are [CONFIRM].

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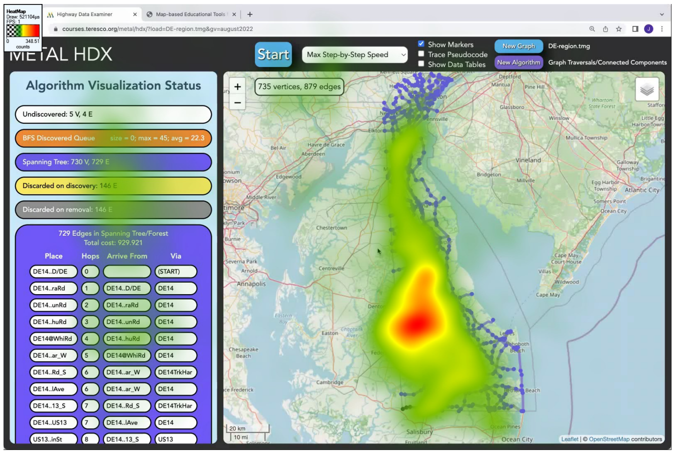


Fig. 1: The METAL HDX interface running a BFS traversal on the Delaware (DE) road network. The left panels show the algorithm-visualization status (including the BFS discovered queue and partial spanning tree) and the per-edge traversal table; the right panel shows the geospatial map. Warm colors overlaid on the map denote aggregated gaze density across Group 1 participants (darker red = more fixations).

3.3 Subjects

Sixty-two participants were recruited at [CONFIRM: institution]; the study was approved under IRB protocol [CONFIRM: number]. Gaze was recorded with a Tobii eye tracker sampled at 60 Hz as participants viewed animated BFS and DFS traversals in the METAL AV system.

We assigned each participant to one of three prior-knowledge groups, according to [CONFIRM: measurement method]:

- **G1 — novice:** no background in computer science.
- **G2 — intermediate:** brief exposure to computer science (e.g., preceding module of coursework).
- **G3 — advanced:** multiple years of formal computer-science coursework.

Every participant viewed both the BFS and DFS animations. With exclusions for quality control, the analyzable set comprised 117 trials (n=59 BFS, n=58 DFS), with counts per group for analyzable set G1=19, G2=21, G3=20. All the results reported below are conditioned on the DE stimulus rendered in the METAL style; the IRN stimulus present in the aggregate dataset is not included in this analysis.

3.4 Stimuli and AOIs

Each trial consisted of a split-screen METAL stimulus: a Pseudocode panel on the left, showing the algorithm's current step, and a geospatial Map on the right, showing what nodes and edges light up as the algorithm traverses through them. Two AOIs were defined on the DE stimulus—Pseudocode and Map. All participants viewed the BFS animation (`DE-bft.wmv`) and the DFS animation (`DE-dft.wmv`); length of each trial was [CONFIRM: number].

For each AOI we made two primary measurements—Fixation Count (FC), and Total Fixation Duration (TFD) across all fixations in that AOI. We derived from these the ratio $\mathrm{TFD}_{\mathrm{pseudo}}/\mathrm{TFD}_{\mathrm{map}}$ of TFD for Pseudocode to TFD for Map, scanner index $\mathrm{VC}_{\mathrm{pseudo}}/\mathrm{TFD}_{\mathrm{pseudo}}$, average depth $\mathrm{TFD}_{\mathrm{pseudo}}/\mathrm{FC}_{\mathrm{pseudo}}$, and switching rate $((\mathrm{VC}_{\mathrm{pseudo}}+\mathrm{VC}_{\mathrm{map}})/(\mathrm{TFD}_{\mathrm{pseudo}}+\mathrm{TFD}_{\mathrm{map}}))$.

3.5 Procedure

Participants completed the study individually in a single session conducted at [CONFIRM: location] between October and November 2022. After signing an informed consent form, each participant was seated in front of the eye-tracking display and completed a [CONFIRM: calibration procedure]. Then, participants viewed both their BFS and DFS animations [CONFIRM: order]—each shown once, per participant. Either as part of, or following this task, we also [CONFIRM: if additional practice trials, comprehension checks, or other post-task questionnaires were administered]. Sessions took approximately [CONFIRM: number] minutes per participant. [CONFIRM: compensation]. We recorded data using [CONFIRM: Tobii software and version] and exported data per-AOI aggregated, for offline analysis.

4 Results

4.1 A Non-Monotonic Pattern of Engagement in Pseudocode

G2 viewed the pseudocode panel longer than G1 and G3. The median pseudocode-to-map ratio was 12.33 in G2. G1 and G3 showed medians of 9.57 and 6.12 respectively. This represents an inverse U relationship with respect to both Total Fixation Duration (TFD) and average fixation depth. Only G1 was different (p=.040) when comparing BFS and DFS. However, there was no difference (p>.50) when comparing G2 and G3. Thus, prior knowledge appears to override the impact of visual organization in determining where participants allocate their gaze. The same shape persists when conditioning on the DFS stimulus alone, with G2 highest and G3 lowest (Figure 2(a)).

4.2 Monotonic Substitution with Respect to Expertise

As expertise increased, the Spearman rank order correlation coefficient (ρ) between pseudocode view TFD and map view TFD became more negative. ρ(G1)=−.034, NS, n=19; ρ(G2)=−.457, p<.050, n=21; ρ(G3)=−.631, p=.003, n=20. Fisher z supported the end point comparison (G1 vs G3, p<.050); however, neither of the middle comparisons reached statistical significance (G1 vs G2 p=.180; G2 vs G3 p=.470).

Similarly, the rank order correlation between pseudocode view fixation count and map view fixation count followed the same trend but had lower values. ρ(G1)=+.042, NS; ρ(G2)=+.326, NS; ρ(G3)=−.161, NS. All three correlations failed to reach statistical significance (p>.100). Similarly, pairwise Fisher z on FC failed to reach significance (p>.10): G2 vs G3 (p=.130); G1 vs G3 (p=.550); G1 vs G2 (p=.380). Partial Spearman controlling for group and algorithm yielded directional relationships similar to those found for TFD. Figure 2 shows both the inverted-U and the lead TFD scatter.

The TFD findings match the predictions of ERE regarding visual search behavior. Novices use both panels as separate units of information processing space; intermediates begin to substitute one unit for another; and experts substitute most heavily. As a result, time devoted to viewing one unit becomes time that is subtracted from viewing another unit. The predicted coordination effect of CTML due to the dual channel nature of the layout is absent in every group's dwell time measures.

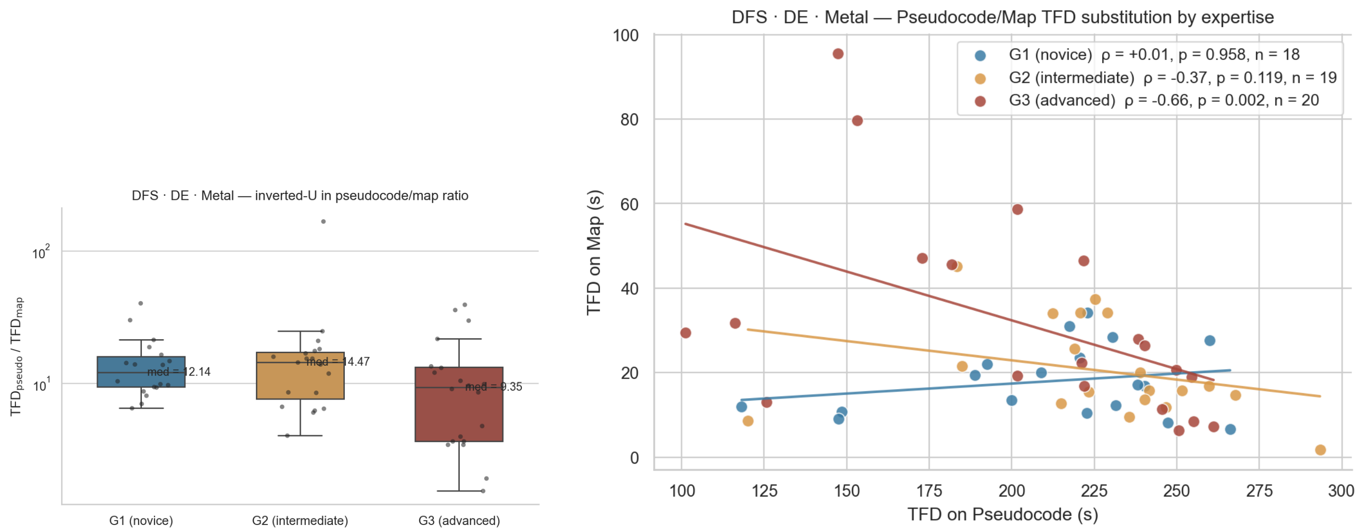
4.3 Attentional Coupling Increases with Increasing Expertise

Attentional coupling increases with increasing expertise. The mean absolute value of Spearman's rank order correlation coefficient (ρ) across all possible metric pairs: G1=.294; G2=.330; G3=.369. Participants without prior experience with visual representations of algorithms demonstrated the least coupled metric pairs. For example, the ratio to scanner index pair ranged from r=−.04 in G1 to r=−.67 in G3. Restricted to the DFS Metal corpus the gradient is sharper still (mean $|\rho|$ = .368, .483, .605 for G1, G2, G3). Scanner index, fixation depth, and switching rate did not differ by group (all p>.20), ruling out a global pacing explanation.

5 Discussion

5.1 Mechanisms

The monotonic increase in the magnitude of negative TFD correlation across the three groups is consistent with CLT and ERE: novices without a schema (G1) consume working memory learning both the algorithm and the visualization, so dwell on one panel is unrelated to dwell on the other; intermediates (G2) develop predictable per-panel reading patterns, so dwell begins to trade off; advanced learners (G3)



(a) Inverted-U in pseudocode/map TFD ratio: G2 highest, G3 lowest.

(b) Primary result: TFD substitution. G3 ρ=−.66, p=.002; G1 flat.

Fig. 2: DFS DE Metal results. (a) Per-participant pseudocode/map TFD ratio shows the inverted-U with G2 at the apex. (b) Per-participant TFD on pseudocode vs. TFD on map: as expertise rises the regression slope rotates from flat (G1) to strongly negative (G3, ρ=−.66, p=.002).

treat each panel as a sufficient cue and substitute one for the other. The inverted-U in dwell ratio (Section 4.1) and the monotonic substitution gradient (Section 4.2) thus describe the same process at the population and per-participant scales respectively.

5.2 Design Implications

Since substitution varies with respect to expertise level, each group of learners uses the split-panel design differently than others. G1: Both panels are used independently. Scaffolding (e.g., line highlighting, annotation, staged display) may be necessary to facilitate coordination between panels.

G2: Panels are partially substitutable with respect to each other while still being engaged simultaneously. This profile fits well with the split-panel format.

G3: The pseudocode panel serves as a verification check for either representation alone.

Metrics related to attention aggregated across all levels of expertise provide average values that characterize none of the groups accurately.

5.3 Limitations

It is expected that roughly five of the 105 Fisher z tests would reach p<.05 by chance alone. While the main TFD contrast (G1 vs G3, p=.04) survives that threshold with some modest correction applied, neither of the FC sign reversals reached statistical significance at the participant level. Learner expertise groups were not randomly assigned, thus group-level differences may be reflective of either general cognitive ability or learner motivation. All visualizations utilized Metal style. Group sizes of 19 to 21 limited power for detecting small effects.

5.4 Conclusion

Time allocated to viewing pseudocode versus maps trades off positively with programming expertise level. Time allocations are essentially independent for novice programmers; intermediate and advanced programmers substitute one for the other increasingly with greater expertise level. Layouts validated at one level of expertise may not generalize; aggregating attentional metrics across levels of expertise hides this substitution gradient.

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